

Data Cleaning, Transformation and Modelling

Theoretical Questions

Q1. What is Data Loading in Power BI and why is it considered the first step of analysis?

Data Loading is the process of connecting Power BI to a data source (Excel, CSV, SQL Server, web, API, etc.) and bringing that data into the Power BI environment (Power Query / the Data model) so it becomes available for cleaning, transformation, and report building.

It is considered the first step of analysis for these reasons:

- No analysis, visualization, or DAX calculation can happen until the raw data physically exists inside Power BI's model.
- Loading determines the shape and quality of the data available downstream — every later step (cleaning, relationships, measures, visuals) depends on it.
- It lets Power BI infer data types, column names, and structure, which is required before transformation rules can be applied.
- Correct loading (choosing the right source, connection mode, and preview) prevents downstream errors such as wrong data types or missing rows.

Q2. Explain the difference between “Load” and “Transform Data” in Power BI.

Aspect	Load	Transform Data
Purpose	Brings the data as-is into the Power BI data model	Opens Power Query Editor to clean/reshape data before it is loaded
When it happens	Final step — data becomes usable in reports	Intermediate step — happens before the final load
Editable?	Data is loaded directly; no changes applied	Full editing: remove duplicates, filter rows, split columns, change types, merge/append queries
Use case	When the source data is already clean and ready	When data has nulls, duplicates, wrong formats, or needs new columns
Engine	Uses the Power BI data engine (VertiQ / xVelocity)	Uses the Power Query (M language) engine

In short: Load = bring the data in; Transform Data = clean/reshape the data in Power Query before it is loaded into the model.

Q3. What is a Fact Table and a Dimension Table? Give examples from the dataset.

A Fact Table stores quantitative, measurable data (numeric facts) generated by business/real-world events, along with foreign keys linking to dimension tables. In this dataset, the daily case-record table containing Confirmed, Recovered, Deaths, and the calculated Recovery Rate for each Date/Country/State combination is the Fact Table — it holds the numbers we aggregate and analyze.

A Dimension Table stores descriptive, textual attributes that give context to the facts — the “who, what, where, when.” In this dataset:

- Dim_Country — Country, Region/Continent
- Dim_State — State, Country (parent)
- Dim_Date — Date, Day, Month, Year, Quarter (a date table)

- Dim_Vaccination_Status — the distinct vaccination categories (Vaccinated / Partially Vaccinated / Not Vaccinated)

The Fact table (Confirmed, Recovered, Deaths, Recovery Rate) connects to these dimension tables through keys such as Country, State, and Date, enabling slicing and filtering in Power BI visuals.

Q4. Why is Star Schema preferred over Snowflake Schema in Power BI?

A Star Schema has one central fact table directly connected to flat, denormalized dimension tables (no sub-levels), whereas a Snowflake Schema normalizes dimensions further into multiple related sub-tables. Power BI performs better with a Star Schema for these reasons:

- Faster query performance — fewer joins means the VertiPaq engine scans and aggregates data more quickly.
- Simpler DAX — measures and relationships are easier to write and maintain with a flat dimension structure.
- Better compression — VertiPaq's columnar storage compresses wide, flat dimension tables very efficiently.
- Easier for report users — filters and slicers behave predictably with a single hop from dimension to fact.
- Recommended by Microsoft — Power BI's own best-practice guidance favors star schemas for both performance and manageability.

Snowflake schemas add extra joins/normalization layers that increase query complexity and slow down DAX calculations, which is why Star Schema is the preferred modelling approach in Power BI.

Practical Questions

Sample Raw Data (before cleaning)

Date	Country	State	Confirmed	Recovered	Deaths	Vaccination_Status
2020-03-01	India	Maharashtra	10	2	0	Not Vaccinated
2020-03-01	India	Maharashtra	10	2	0	Not Vaccinated
2020-03-02	India	Kerala	5	1	0	(blank)
2020-03-02	USA	California	20	5	1	Partially Vaccinated
2020-03-02	USA	New York	15	3	0	Fully Vaccinated
2020-03-03	India	Maharashtra	12	4	0	(blank)
2020-03-03	USA	California	25	8	1	Fully Vaccinated
2020-03-04	India	Kerala	7	2	0	Partially Vaccinated
2020-03-04	USA	New York	18	6	0	(blank)
2020-03-04	India	Maharashtra	12	4	0	Not Vaccinated

Q5. Identify and remove duplicate records based on Date, Country, and State.

Steps in Power BI (Power Query Editor):

- Home → Transform Data to open Power Query Editor.
- Select the columns Date, Country, and State together (Ctrl + click each header).
- Home → Remove Rows → Remove Duplicates (this keeps the first occurrence of each Date/Country/State combination and deletes the rest).

Equivalent Power Query M code:

```
let
    Source = Csv.Document(File.Contents("Corona_Virus_2020.csv")),
    PromotedHeaders = Table.PromoteHeaders(Source),
    RemovedDuplicates = Table.Distinct(PromotedHeaders,
        {"Date", "Country", "State"})
in
    RemovedDuplicates
```

Result: Row 2 in the sample (2020-03-01, India, Maharashtra) was an exact duplicate of Row 1 on Date/Country/State and was removed. The table went from 10 rows to 9 rows.

Q6. Identify and replace null values in Vaccination_Status.

Steps in Power BI:

- In Power Query, right-click the Vaccination_Status column header → Replace Values, or use the column's filter dropdown to spot (blank)/null rows first.
- Decide the replacement rule: a fixed placeholder like "Unknown", or the most frequent (mode) category, or 0/blank-preserving logic depending on business need. Here the most frequent category (mode) — "Fully Vaccinated" — was used to fill the nulls, a common statistical approach for categorical data.
- Home → Replace Values → set "Value To Find" = null, "Replace With" = the chosen value (or use the Fill Down feature if nulls should inherit the previous row's status).

Equivalent Power Query M code:

```
let
    ReplacedNulls = Table.ReplaceValue(RemovedDuplicates,
        null, "Fully Vaccinated", Replacer.ReplaceValue,
        {"Vaccination_Status"})
in
    ReplacedNulls
```

Result: The two blank Vaccination_Status cells (rows for Kerala 03-02 and Maharashtra 03-03) were replaced with "Fully Vaccinated", the most frequent category found in the column.

Q7. Create a new column to calculate Recovery Rate.

Recovery Rate is calculated as: $\text{Recovery Rate (\%)} = (\text{Recovered} / \text{Confirmed}) \times 100$

Option A — Power Query (Add Column → Custom Column):

```
let
    AddedRecoveryRate = Table.AddColumn(ReplacedNulls,
        "Recovery_Rate (%)", each
        if [Confirmed] = 0 then 0
        else Number.Round([Recovered] / [Confirmed] * 100, 2))
in
    AddedRecoveryRate
```

Option B — DAX Calculated Column (in the Data/Model view):

```
Recovery Rate (%) =
DIVIDE ( Corona_Virus_2020[Recovered], Corona_Virus_2020[Confirmed], 0 ) * 100
```

DIVIDE() is preferred over a plain "/" in DAX because it safely returns 0 (or a chosen alternate value) instead of an error when Confirmed = 0.

Q8. Create a summarized table showing total confirmed cases by Country.

Steps in Power BI:

- Power Query method: select the table → Transform → Group By → Group by "Country", New column name "Total_Confirmed", Operation = Sum, Column = Confirmed.
- Report-view method: drag Country into a Table/Matrix visual, then drag Confirmed into the same visual — Power BI sums it automatically by default.

Equivalent Power Query M code:

```
let
    GroupedByCountry = Table.Group(AddedRecoveryRate,
        {"Country"},
        {"Total_Confirmed", each List.Sum([Confirmed]), type number})
in
    GroupedByCountry
```

Equivalent DAX Measure (for use in cards/visuals instead of a static table):

```
Total Confirmed = SUM ( Corona_Virus_2020[Confirmed] )
```

Final Cleaned Table (output of Q5–Q7)

Date	Country	State	Confirmed	Recovered	Deaths	Vaccination_Status	Recovery_Rate(%)
2020-03-01	India	Maharashtra	10	2	0	Not Vaccinated	20.00
2020-03-02	India	Kerala	5	1	0	Fully Vaccinated	20.00
2020-03-02	USA	California	20	5	1	Partially Vaccinated	25.00
2020-03-02	USA	New York	15	3	0	Fully Vaccinated	20.00
2020-03-03	India	Maharashtra	12	4	0	Fully Vaccinated	33.33
2020-03-03	USA	California	25	8	1	Fully Vaccinated	32.00
2020-03-04	India	Kerala	7	2	0	Partially Vaccinated	28.57
2020-03-04	USA	New York	18	6	0	Fully Vaccinated	33.33
2020-03-04	India	Maharashtra	12	4	0	Not Vaccinated	33.33

Summarized Table — Total Confirmed Cases by Country (output of Q8)

Country	Total Confirmed
India	46
USA	78